

LUNDS UNIVERSITET

ETIN71

Introduction to Integrated Electronics

Design and Evaluation of a Nanowire CMOS Differential Amplifier

Project Report

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1 Introduction

This report details the design, simulation and evaluation of a two-stage differential amplifier shown in Figure 1. The circuit was designed in Qucs [1] using nanowire (NW) MOSFET models. These complementary NW CMOS devices feature a gate-all-around geometry and utilize III-V materials (such as InAs) to achieve higher carrier mobility and improved electrostatics compared to traditional planar transistors [2]. Because their behavior is decoupled from traditional area scaling, they can be accurately modeled and evaluated much like standard short-channel MOSFETs.

Throughout this project, the provided Nanoelectronics Student Component Library (NSCL) was used to construct and systematically evaluate a complete analog amplifier circuit. The design from Figure 1 incorporates an NMOS differential input pair (Q_1, Q_2), a PMOS current mirror active load (Q_3, Q_4), a current steering bias network (Q_6, Q_7, Q_8), and a common-drain output stage (Q_5) acting as a voltage follower. By iteratively tuning component geometries through DC bias configurations and evaluating the frequency response via AC simulations, the project aimed to verify the real-world capabilities of the NW MOSFET technology and compare the resulting circuit's performance against the theoretical characteristics of an ideal operational amplifier (op-amp).

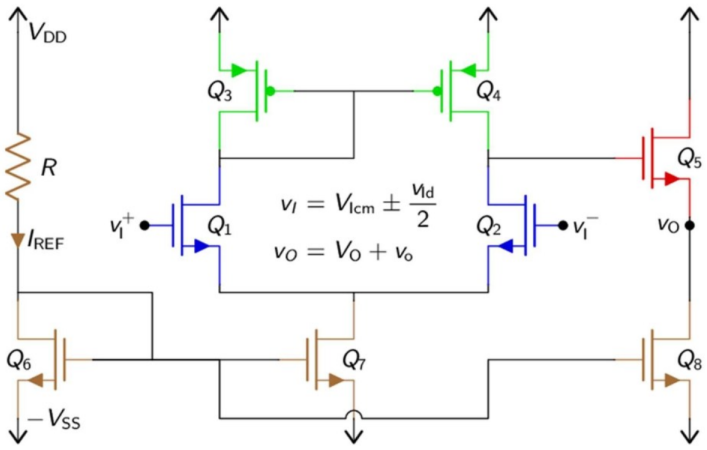


Figure 1: Differential Amplifier provided by the lab manual [2] comprised of an NMOS differential input (blue), PMOS current mirror load (green), NMOS current steering bias (yellow) and an NMOS common drain output (red).

2 Theory

This section will explore the characteristics of an ideal op-amp, the important MOSFET parameters related to these ideal characteristics, and MOSFET configurations used in our amplifier design.

2.1 Ideal Operational Amplifier Characteristics

An op-amp takes two voltage inputs, one non-inverting, v_i^+, and one inverting, v_i^-, amplifies their difference, and outputs a voltage to be used by the remaining circuit, v_o. The relationship between the differential inputs and output voltage is given by:

$$v_o = A_{OL}(v_i^+ - v_i^-) \tag{1}$$

where A_{OL} is the open loop amplifier gain, which determines how effectively the op-amp increases the signal. An open loop (OL) op-amp has no feedback connection from its output to either input. For an ideal OL op-amp, $A_{OL} = \infty$.

The ideal op-amp also assumes infinite input impedance $R_{in} = \infty$ so that no current is drawn by the input terminals and therefore the op-amp does not load the signal source. Similarly we assume $R_{out} = 0$ so the output voltage is unaffected by the current drawn by the load.

For equal input voltages, the differential input is zero, and therefore we expect $v_o = 0$. This is not always the case due to internal device mismatching.

Another common property is gain performance across frequencies. The upper cutoff frequency is the frequency at which the low frequency gain decreases by 3 dB. For this low-pass response, the bandwidth (BW) extends from DC to the upper cutoff frequency. For an ideal op-amp with no capacitive effects, $BW = \infty$.

A voltage follower is a closed loop (CL) configuration, where the output voltage is connected directly to the inverting input resulting in $v_o = v_i^-$. Plugging this condition into Equation (1) we can find the closed loop gain, A_{CL}:

$$\begin{aligned} v_o &= A_{OL}\left(v_i^+ - v_o\right) \\ v_o\left(1 + A_{OL}\right) &= A_{OL} v_i^+ \\ A_{CL} = \frac{v_o}{v_i^+} &= \frac{A_{OL}}{1 + A_{OL}} \end{aligned} \tag{2}$$

As A_{OL} approaches infinity, A_{CL} approaches 1. This is called the voltage follower configuration since v_o approaches v_i^+ as A_{OL} increases.

2.2 Nanowire MOSFET Large & Small Signal Models

A MOSFET is a voltage-controlled current source, whereby supplying specific voltages to each of its three terminals (drain, gate, source), we can influence the drain current, I_D, and its mode of operation. For amplifier design, we want all MOSFETs to operate in saturation. For an NMOS transistor, this is satisfied by the conditions:

$$V_{GS} > V_t, \qquad V_{DS} \geq V_{GS} - V_t \tag{3}$$

For a PMOS transistor, the analogous conditions are expressed using V_{SG}, V_{SD} and $|V_{tp}|$.

The I_D relationship for a MOSFET in saturation is approximately given by:

$$I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{GS} - V_t)^2 (1 + \lambda V_{DS}) \tag{4}$$

where k_n' is a material parameter, $\frac{W}{L}$ is the width-to-channel-length ratio, V_{GS} is the potential difference between the gate and source terminals, V_t is the threshold voltage where a channel begins to form and the $1 + \lambda V_{DS}$ describes channel length modulation. This is an approximation since the NSCL MOSFET performance depends on operating points and other short-channel effects [2]. For the nanowire MOSFETs, W is determined by the following relation with $W_{\text{eq}} = 100\,\text{nm}$ and NW being the number of nanowires, the parameter we used to design our amplifier:

$$W = W_{\text{eq}} N W \tag{5}$$

For our analysis, we consider a DC voltage operating point Q and small variations about Q. The small-signal change in current, Δi_d, due to a small gate-source voltage change, Δv_{gs}, can be described by a linear small-signal model shown in Figure 2 and by Equation (7).

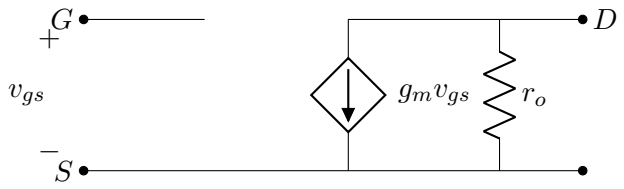


Figure 2: Simplified small-signal model of an NMOS transistor

$$\Delta i_{d|Q} = g_m \Delta v_{gs|Q} \tag{6}$$

$$g_m = \left.\frac{\partial I_D}{\partial V_{GS}}\right|_Q \tag{7}$$

where g_m is the transconductance determined by material properties and DC operating point values. This shows that a small change in the gate-source voltage produces a proportional small-signal change in the drain current, resembling an amplifier.

In Figure 2, notice there is also an output resistance r_o. This comes from channel length modulation. As V_{DS} increases while the MOSFET remains in saturation, I_D also increases slightly.

$$r_o = \left.\frac{\partial V_{DS}}{\partial I_D}\right|_{V_{GS},Q} = \left(\left.\frac{\partial I_D}{\partial V_{DS}}\right|_{V_{GS},Q}\right)^{-1} \tag{8}$$

In a common-source stage, the small-signal drain current produces an output voltage $\Delta v_o = -\Delta i_d r_o$. Combining this with Equation (6) gives the intrinsic voltage gain

$$A_{v,\text{int}} = \frac{\Delta v_o}{\Delta v_{gs}} \approx -g_m r_o, \qquad |A_{v,\text{int}}| \approx g_m r_o. \tag{9}$$

Using the saturation model in Equation (4), g_m and r_o can be approximated by

$$g_m \approx \frac{2I_D}{V_{OV}}, \qquad r_o \approx \frac{1}{\lambda I_D}, \qquad V_{OV} = V_{GS} - V_t. \tag{10}$$

For a fixed transistor size, $I_D \propto V_{OV}^2$, giving

$$g_m \propto \sqrt{I_D}, \qquad r_o \propto \frac{1}{I_D}, \qquad g_m r_o \propto \frac{1}{\sqrt{I_D}}. \tag{11}$$

These relationships are approximations for the short-channel NSCL devices but show how increasing drain current can increase g_m while reducing r_o and intrinsic gain.

2.3 Amplifier Building Blocks

2.3.1 Current Steering

Q_6, Q_7 and Q_8 form a current mirror that establishes the amplifier bias currents. Q_6 is diode connected ($V_G = V_D$) and carries I_{REF}. Since Q_7 and Q_8 share its gate voltage and channel length, using Equation 4, their currents are approximately scaled by their number of nanowires:

$$\frac{I_{D7}}{I_{REF}} \approx \frac{NW_7}{NW_6}, \qquad \frac{I_{D8}}{I_{REF}} \approx \frac{NW_8}{NW_6}. \tag{12}$$

Q_7 supplies the differential-pair tail current, while Q_8 biases the output stage. The scaling is approximate because unequal V_{DS} values and finite output resistance affect the mirrored currents.

2.3.2 Differential Pair

Q_1 and Q_2 form an NMOS differential pair driven by $v_{id} = v_i^+ - v_i^-$. When $v_{id} = 0$, matched devices divide the Q_7 tail current approximately equally, so $I_{D1} \approx I_{D2} \approx I_{D7}/2$. For a small positive v_{id}, I_{D1} increases and I_{D2} decreases by Equation 6.

$$\Delta i_{d1} = +\frac{g_m}{2}\Delta v_{id}, \qquad \Delta i_{d2} \ = -\frac{g_m}{2}\Delta v_{id}. \tag{13}$$

The stage therefore converts the differential input voltage into opposing drain-current changes.

2.3.3 Current Mirror Load

Q_3 and Q_4 form a PMOS current-mirror active load. Q_3 is diode connected, and Q_4 approximately mirrors its current into the Q_2/Q_4 node. When v_i^+ increases relative to v_i^-, Q_4 mirrors the increased Q_1 current described by Equation 13 into the output-side branch. Together, Q_1–Q_4 form the first gain stage. The resistance at the shared drain node of Q_2 and Q_4 is approximately $r_{o2} \parallel r_{o4}$, which is a high resistance because both transistors operate in saturation. The mirror converts the differential current into a voltage with a gain A_{v1}:

$$A_{v1} \approx g_m \left(r_{o2} \parallel r_{o4}\right). \tag{14}$$

where g_m is the transconductance of Q_1/Q_2 assuming matching.

2.3.4 Common Drain Output Stage

Q_5 is a common-drain amplifier, or source follower, driven by the output of the first gain stage and biased by the Q_8 current sink. The small-signal resistance at its source is approximately $R_S = r_{o5} \parallel r_{o8}$. Neglecting body effect, its voltage gain is

$$A_{v2} \approx \frac{g_{m5}R_S}{1 + g_{m5}R_S}. \tag{15}$$

When $g_{m5}R_S \gg 1$, this gain approaches unity, so the output voltage follows the gate voltage. The source follower transfers the voltage developed by the first gain stage to the output with a gain close to unity. Since the gate of Q_5 draws negligible current from the Q_2/Q_4 node ($I_{G5} \approx 0$), it does not significantly affect the first-stage gain. Its lower output resistance of approximately $1/g_{m5} \parallel r_{o5} \parallel r_{o8}$ reduces the change in output voltage when a load is connected, resembling ideal op-amp characteristics.

3 Method

To design and test the differential amplifier, Qucs-S [1] was used with NMOS and PMOS components supplied by NSCL. The analysis consisted of designing the differential amplifier, testing it in open-loop (OL) and closed-loop (CL) configurations, and characterizing the NMOS and PMOS devices using parameters selected in the design.

3.1 Differential Amplifier Design

The circuit shown in Figure 1 was constructed using the NSCL models. Q_1, Q_2, and Q_5–Q_8 were NMOS devices, while Q_3 and Q_4 were PMOS devices. The NW and L parameters were exposed and controlled using a Qucs-S parameter block. The fixed design values are given in Table 1. During initial sizing, resistor R was replaced by a 1 mA current source to establish I_{REF}.

Parameter	Value
V_{DD}, V_{SS}	1 V, −1 V
v_i^+, v_i^-	0 V, 0 V
I_{REF}	1 mA
L_1–L_5	20 nm
L_6–L_8	50 nm
NW_1, NW_2	50, 50
NW_6	20

Table 1: Fixed values used during differential amplifier design.

DC operating-point simulations were performed while NW_7 and NW_8 were adjusted until Q_1–Q_5 each carried approximately 10 mA. NW_3 and NW_4 were adjusted until $|V_{DS}|$ was approximately equal across Q_1–Q_4, while NW_5 was adjusted until the output voltage was $v_o = 0$ V. These parameters were adjusted iteratively until the three conditions were approximately satisfied and changing any value by a single nanowire increased the corresponding deviation from its target. We also verified that each transistor was operating in saturation.

After sizing, the gate voltage of Q_6 was used to determine the value of resistor R required to replace the current source and supply 1 mA to the NMOS current mirror.

3.2 OL and CL Configurations

The amplifier subcircuit was supplied with $V_{DD} = 1$ V and $V_{SS} = -1$ V. For the OL configuration, a 10 mV AC source was connected to v_i^+ and v_i^- was grounded. A 10 MΩ resistor was connected as an output load. The AC response was simulated using 200 logarithmically spaced points from 20 Hz to 100 GHz. The same setup was used for the CL configuration, with v_o connected directly to v_i^-.

3.3 NMOS and PMOS Characterization

The individual NMOS and PMOS test benches supplied by NSCL were used to obtain their transfer and output characteristics. The NMOS values of NW and L were set to those of Q_1, while the PMOS values were set to those of Q_3. The voltage sweeps were selected to pass

through the operating point $V_{DS} = V_{GS} = 0.6\,\mathrm{V}$ for the NMOS and $V_{DS} = V_{GS} = -0.6\,\mathrm{V}$ for the PMOS. The resulting curves were used to determine the small-signal parameters at these operating points.

4 Results

4.1 Differential Amplifier Design

The final design parameters and corresponding DC operating point are shown in Table 2. The drain currents were measured using ammeters in Qucs-S, while the drain-source and output voltages were measured using wire labels.

Parameter	Final value
$NW_3 = NW_4$	164
NW_5	202
NW_7	380
NW_8	118
$I_{D1} = I_{D2} = I_{D3} = I_{D4}$	$9.84\,\mathrm{mA}$
I_{D5}	$10.1\,\mathrm{mA}$
V_{DS1}	$0.844\,\mathrm{V}$
$\lvert V_{DS3}\rvert$	$0.847\,\mathrm{V}$
$V_{DS1} - \lvert V_{DS3}\rvert$	$-2.74\,\mathrm{mV}$
v_o	$0.225\,\mathrm{mV}$

Table 2: Final differential amplifier parameters and DC operating point. The displayed V_{DS} values are rounded to three decimal places by Qucs-S, while the drain-source difference was calculated using the unrounded simulation values.

After sizing, the gate voltage of Q_6 was measured as $V_{G6} = -0.709\,\mathrm{V}$. The resistance required to maintain $I_{REF} = 1\,\mathrm{mA}$ was calculated using Ohm's law:

$$R = \frac{V_{DD} - V_{G6}}{I_{REF}} = \frac{1 - (-0.709)}{1\,\mathrm{mA}} = 1.709\,\mathrm{k\Omega}. \tag{16}$$

The temporary current source was replaced by this resistor. The resulting changes in the drain-current error, V_{DS} mismatch and output offset were smaller than their existing deviations from the target values and were therefore treated as negligible. The circuit was then saved as a subcircuit with labelled supply, input and output ports.

4.2 Differential Amplifier Performance in OL and CL

The OL and CL gain responses of the final design are shown in Figure 3. The low-frequency OL gain was $A_{OL} = 10.2\,\mathrm{dB}$. Linear interpolation between two points around the $-3\,\mathrm{dB}$ point gave an OL bandwidth of $BW_{OL} = 11.93\,\mathrm{GHz}$.

The low-frequency CL gain was $A_{CL} = -3.27\,\mathrm{dB}$. Similarly to the OL case, linear interpolation gave a CL bandwidth of $BW_{CL} = 45.57\,\mathrm{GHz}$. Simulating an ideal op-amp with current input gain $G_{ms} = 100$, voltage output gain $R_{mo} = 100$ and no capacitances produced $A_{OL} = 79.2\,\mathrm{dB}$, $A_{CL} = 0\,\mathrm{dB}$, and infinite bandwidths for both.

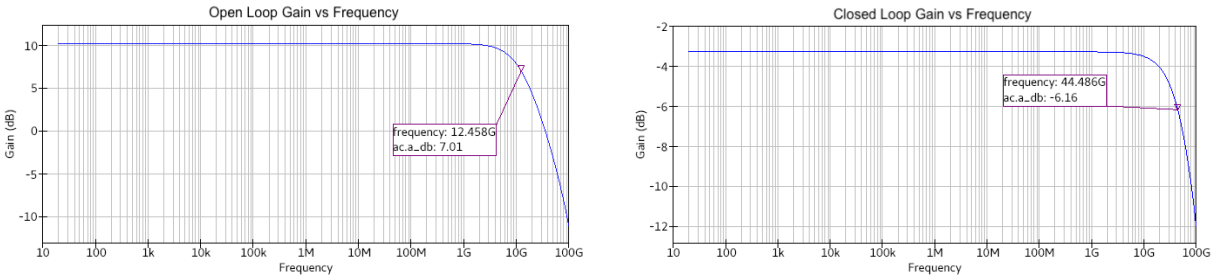


Figure 3: Open and closed loop Gain (dB) plotted against frequency. Frequency is plotted logarithmically ranging from 20 Hz to 100 GHz. The marker indicates a value used to interpolate the bandwidth.

4.3 Low-Frequency Small-Signal Parameters

From Equation (7), g_m can be found by measuring the slope using two markers about the $V_{DS} = V_{GS} = Q$ operating point on the I_{DS}, V_{GS} curve. From Equation (8), r_o can be found by similarly measuring the slope's inverse on the I_{DS}, V_{DS} curve about Q. These results are presented in Table 3.

	r_o (Ω)	g_m (mS)
NMOS	500	13.3
PMOS	500	11.0

Table 3: Small-signal parameters extracted from the DC output and transfer characteristics of the NSCL-provided NMOS and PMOS components operating with the same parameters as Q_1 from Table 1 and Q_3 from Table 2.

5 Discussion

5.1 DC Operating Point and Sizing

The current-steering network scaled the 1 mA reference current using the nanowire ratios. For $NW_7 = 380$ and $NW_6 = 20$, Equation (12) predicts

$$I_{D7} \approx 1\,\text{mA}\frac{380}{20} = 19\,\text{mA}. \tag{17}$$

The matched differential pair should therefore carry approximately 9.5 mA per branch, which is close to the simulated 9.84 mA through Q_1–Q_4. Since Q_8 supplies only the output branch, its required current and NW_8 are smaller than those of Q_7. The exact current ratios differ from the ideal width scaling because Equation (12) assumes equal drain-source voltages and neglects other short-channel effects.

The current steering network sets the currents $I_{D1} = I_{D2} = I_{D3} = I_{D4}$ as seen in Table 2. As a result, when increasing NW_3 and NW_4 while maintaining the same I_D, V_{SG} must decrease for each PMOS by Equation (4). Since $V_{S3} = V_{DD}$ is fixed, $V_{G3} = V_{D3}$ must increase. Therefore V_{SD3} decreases when increasing NW_3, and V_{DS1} increases. This opposing change allows us to tune the drain-source voltages for Q_1 to Q_4.

Similarly, Q_5's drain current is set by Q_8. Increasing NW_5 reduces the V_{GS5} required to maintain the same drain current. Since V_{G5} is set by the current-mirror load, $V_{S5} = v_o$ therefore increases.

This allows us to tune the output voltage. The final value of $v_o = 0.225\,\text{mV}$ is a small non-ideality because an ideal amplifier would produce zero output for equal input voltages.

5.2 Gain and Feedback Performance

The finite open-loop gain can be estimated from the measured small-signal parameters. Using Equation (14),

$$A_{v1} \approx 13.3\,\text{mS}\,(500\,\Omega \parallel 500\,\Omega) = 3.33, \tag{18}$$

which corresponds to 10.4 dB. This is close to the simulated open-loop gain of 10.2 dB. The small difference is consistent with the source-follower gain being below unity and with the approximations made in the small-signal model. The finite values of r_{o2} and r_{o4} therefore limited the voltage gain compared with an ideal op-amp. As discussed in Section 2.3.4, the output resistance is small but non-zero, unlike that of an ideal op-amp. Therefore, different loads which draw different currents will affect the voltage delivered to the load.

The open-loop gain could be increased by reducing I_{D7}, which would reduce the currents through Q_1–Q_4. For fixed transistor sizes, Equation (11) shows that $g_m r_o \propto 1/\sqrt{I_D}$. Although reducing the drain current decreases g_m, it increases r_o more strongly and therefore increases the intrinsic voltage gain. This would, however, move the drain currents away from the specified 10 mA operating condition. Additionally, increasing NW_1 and NW_2 could increase g_m and therefore increase the OL gain, although the resulting operating point would need to be retuned.

Using Equation (2) and the experimental open-loop gain,

$$A_{CL} = \frac{3.24}{1 + 3.24} = 0.764, \tag{19}$$

which corresponds to -2.34 dB, compared with the simulated -3.27 dB. Equation (2) represents the amplifier as an ideal voltage-gain block, while the transistor circuit contains finite output resistances and a non-ideal source follower. The prediction therefore follows the simulated trend but does not give the exact closed-loop gain. Negative feedback still makes the response more closely resemble an ideal voltage follower by driving v_o towards v_i^+.

The parasitic capacitances in the MOSFET models caused the gain to decrease at high frequencies, producing a finite bandwidth instead of the infinite bandwidth of an ideal op-amp. In the CL configuration, any change in v_o also changes the inverting input voltage and therefore opposes the original change. This negative feedback makes the closed-loop response less sensitive to the reduction in open-loop gain with frequency. Consequently, the gain decreased from 10.2 dB in OL to -3.27 dB in CL, while the bandwidth increased from 11.93 GHz to 45.57 GHz. Additionally, these parasitic capacitances lower impedance at higher frequencies. As a result, the input impedance deviates further from the ideal infinite input impedance at high frequencies.

6 Conclusion

Through iterative DC analysis, we successfully tuned NW parameters to establish specified currents from a current steering network, equal drain-source voltages across a differential input and current mirror load, and reduced output offset voltage. AC testbenches produced an $A_{OL} = 10.2$ dB, much smaller than that of the ideal op-amp due to the MOSFETs' finite output resistances and finite transconductance determined by device geometry and operating points. Similarly, $A_{CL} = -3.27$ dB also differed from the ideal 0 dB. Due to parasitic capacitances, the bandwidths of both the OL and CL amplifiers are finite.

References

[1] Qucs-S Project, "Qucs-S: A Qucs GUI for SPICE simulators." `https://ra3xdh.github.io/`

[2] *Simulation Project: ETIN71 – Introduction to Integrated Electronics*, Lund University, 2026.